FML Assignment

WORK FLOW FOR ALL LAB EXPERIMENTS

Exp NO - 1

Perform data preprocessing on a given dataset

1. Load the dataset - Load the titanic dataset using seaborn
2. Select Features - select survived, pclass, sex, age, fare and embarked columns.
3. check Missing values - count the missing values in each column.
4. Handle Missing values - Fill missing age values with the median and missing embarked values with 's'
5. convert categorical Data - convert sex and embarked columns into numerical o/1 columns using pandas.
6. Display cleaned data - Display the first 5 rows of the cleaned dataset.
7. Verify Data - check whether any missing values remain.

Exp NO - 2

Implement dataset splitting and cross-validation

1. Load the dataset - load the iris flower dataset using scikit-learn.

2. Split the data - Divide the data into 80% training data and 20% testing data.

3. Create the Model - create a KNN classifier with (K = 5)

4. Train the Model - Train the model using the training data

5. Evaluate the model - calculate the model accuracy using the testing data.

6. Predict New flower - predict the species of a new flower using its four measurements.

7. Display the result - print the predicted flower species.

Exp NO-3

Implement Linear Regression and evaluate Performance

1. Load the dataset - Load the car mileage dataset using seaborn and remove missing values.

2. Select Features - Select cylinders, displacement, horsepower, weight, acceleration and model year as input features.

3. Select Target - choose mpg as the target variable.

4. Split the data - Divide the data into training and testing sets (80% training and 20% testing)

5. create the model - create a Linear Regression Model.

6. Train the model - Train the model using the training data.

7. Evaluate the model - calculate the R^2 score using the testing data.

8. predict Mileage - Predict the mileage of a new car.

9. Display the result - print the predicted mileage in mpg.

Exp-4

Implement a classification model (KNN/SVM) and compute evaluation metrics.

1. Load the Dataset - Read social_Network_Ads.csv using pandas

2. Select Features - choose Age and Estimated Salary as input features and purchased as the target.

3. Split the data - Divide the data into 75% training data and 25% testing data

4. Scale the data - use standardscaler to Standardize the training and testing data.

5. Create the Models - Create KNN with (K=5) and an SVM classifier.

6. Train the Models - Train both models using the training data.

7. Evaluate the Models - Calculate and display the accuracy of KNN and SVM

8. Predict Purchase - Predict whether a new person will purchase using KNN

9. Display the Result - Print the predicted purchase result.

Exp-5

Implement K-Means clustering

1. Load the Dataset - Read Mall_Customers.csv using pandas.

2. Select Features - Choose Annual Income and Spending Score as input features.

3. Create K-Means Model - Set n_clusters = 5 to divide customers into 5 groups.

4. Train the Model - Apply K-Means to group customers based on similarity.

5. Assign Clusters - Store each customers ~~based on sim~~ group number in the group column.

6. Predict New customer - Identify the group of a new customer using income and spending score.

7. Display the result - print the no. of customers in each group and the new customer's group number.

Exp No - 6

Perform hierarchical clustering and compare results.

1. Load the Dataset - Read Mall_Customers.csv using Pandas.

2. Select Features - choose Annual Income and Spending score as input features.

3. Perform Hierarchical clustering - Divide the customers into 5 groups using hierarchical clustering.

4. Assign clusters - store each customer's group number in the group column.

5. Calculate Group Averages - Find the average income and spending score of each group.

6. Find the closest Group - calculate the distance between a new customer and each group's average.

7. Display the Result - print the closest group for the new customer.

Exp No-7

Implement Random Forest and analyze performance

1. Load the Dataset - Load the Breast cancer dataset using Scikit-learn
2. Split the data - divide the data into 80% training data and 20% testing data.
3. Create the Models - Create a Decision Tree classifier and a Random Forest classifier with 100 trees.
4. Train the Models - Train both models using the training data
5. Evaluate the Models - calculate and display the accuracy of both models using the testing data.
6. Predict New Tumour - predict the tumour type of new sample using the Random Forest model
7. Display the Result - print the predicted tumour type as malignant or benign.

Exp No-8

Implement boosting techniques and compare results

1. Load the Dataset - Read heart.csv using pandas.
2. Select Features - use all columns expect target as input features and target as output
3. Split the data - Divide the data into 80% training data and 20% testing data.
4. Create the model - create a gradient boosting classifier
5. Train the Model - Train the model using the training data.
6. Evaluate the Model - calculate and display the model accuracy using the testing data.
7. Predict Heart Disease - predict whether a new patient has heart disease.
8. Display the result - Print the prediction as 1 (disease) or 0 (no disease)

Exp-No-9

Implement a basic reinforcement learning model

1. Create the Environment - create the 4 x 4 FrozenLake game using Gymnasium.
2. Initialize Q-Table - create a Q-table with

16 states and 4 actions, initialized to zero

3. Train the agent - Run 2000 practice episodes to learn the best actions.

4. choose an Action - Explore with a random action 20% of the time or choose the best-known action.

5. perform the Action - Move to the next state and observe the reward.

6. update the Q-table - update action values using the Q-learning formula.

7. Display the result - print the best move for selected squares.

Exp no - 10

Build a neural network using TensorFlow/Keras

1. Load the Dataset - Load the handwritten digit dataset using scikit-learn.

2. split the data - Divide the data into 80% training data and 20% testing data.

3. create the model - Build a neural network with 64 input features, one hidden layer with 64 neurons, and an output layer with

neurons.

4. compile the model - use the Adam optimizer, sparse categorical cross-entropy loss, and accuracy metric.

5. Train the model - Training the model for 10 epochs using the training data.

6. Evaluate the model - calculate the test accuracy using the testing data.

7. Predict a Digit - predict the digit from a new handwritten digit image.

8. Display the result - print the predicted digit from 0 to 9.